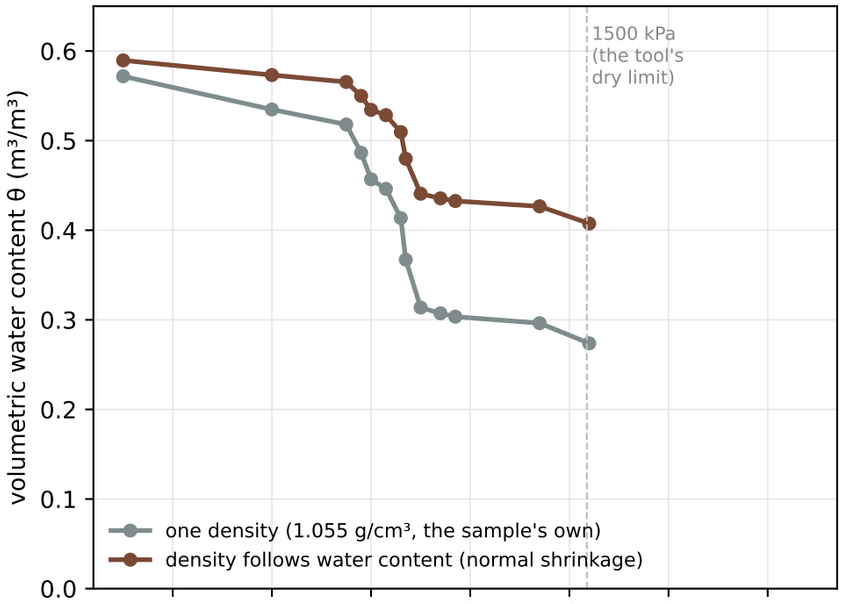
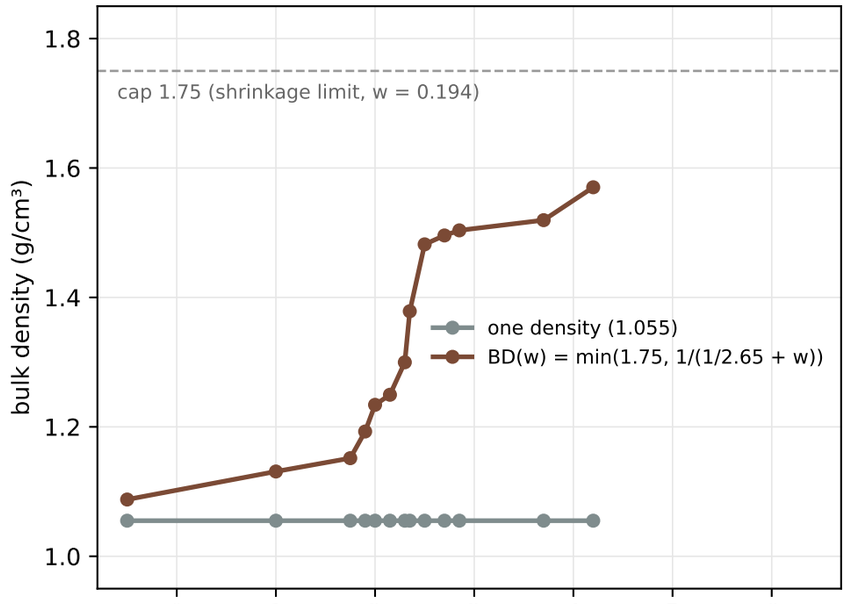
Two vertisols, two regions: converting retention curves with a shrinking bulk density

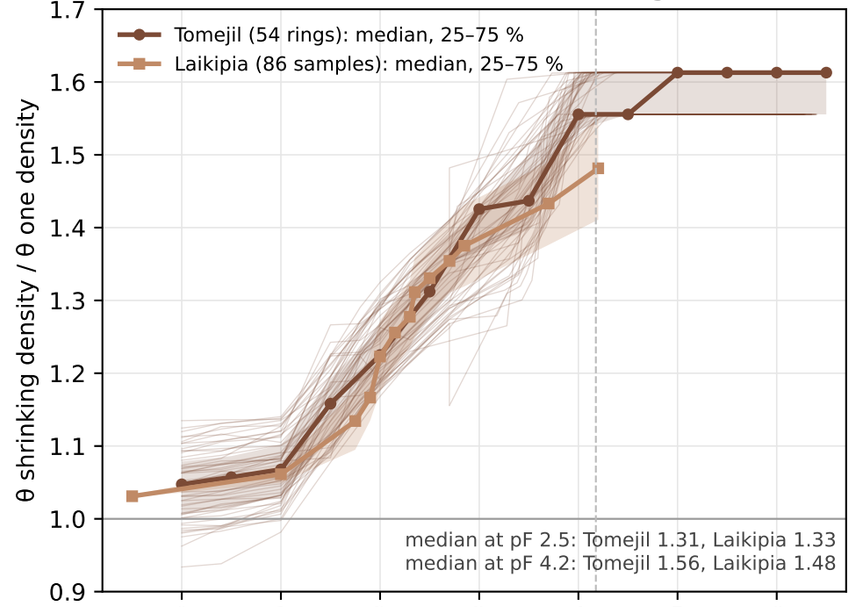
a Laikipia (Kenya), sample WL_3LT07414: retention curve



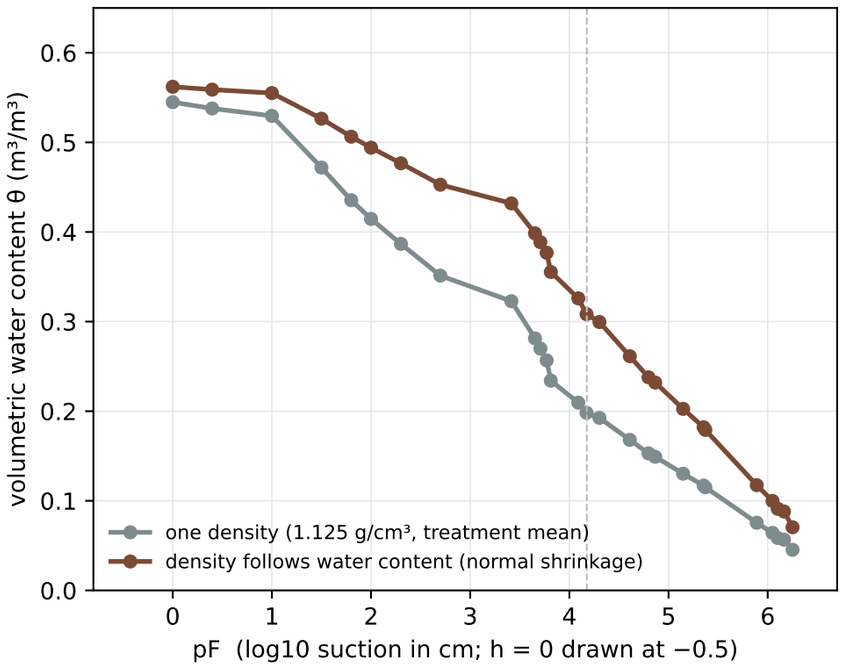
b Laikipia: density used for the conversion



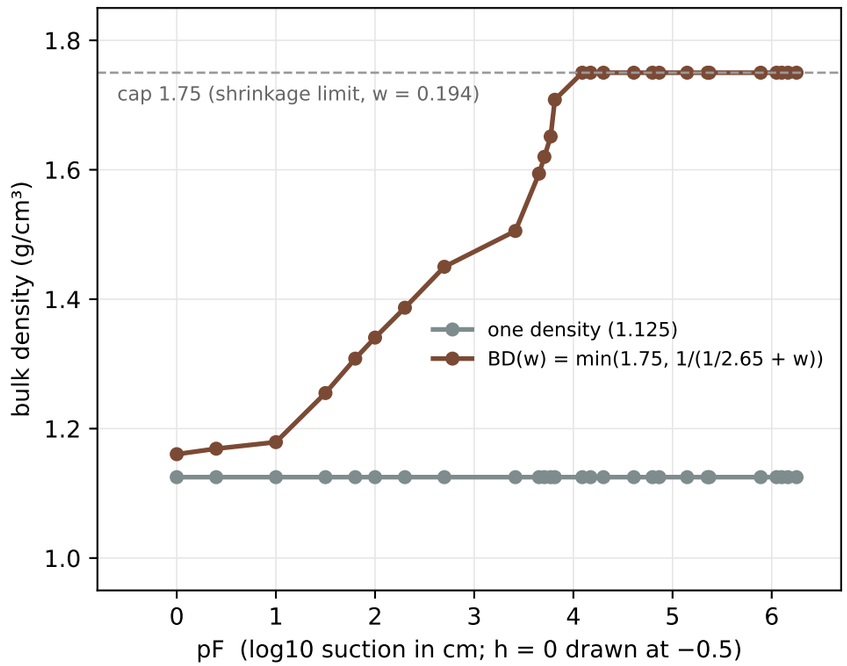
c both soils: how much the conversion changes θ



d Tomejil (Andalusia), ring DD30: retention curve



e Tomejil: density used for the conversion



f the conversion

At every suction:
  w = θ / BD (one density) gravimetric water
  BD(w) = min(1.75, 1 / (1/2.65 + w)) normal shrinkage:
    the soil loses volume equal to the water it loses,
    until the shrinkage limit (BD 1.75, w = 0.194;
    median oven-dry clod density of KSSL Vertisols)
  θ = w · BD(w)

One density:
  Laikipia — each sample's own, behind the corrected porosity
  Tomejil — no ring density; the treatment mean,
    1.085 (conventional tillage) or 1.125 (direct drill)
    (Ordóñez Fernández et al. 2007); curves gravimetric

Both soils start near 1.1–1.2 g/cm³ when saturated, so the conversion lifts the dry end by about half. Beyond pF ~3.8 most Tomejil rings reach the cap, the ratio then cap / treatment density (1.75 / 1.125 = 1.56; 1.75 / 1.085 = 1.61).

The agreement is partly built in (same formula and cap). Independent support: with the converted curves the tool reads both as clay — Tomejil 54 % clay predicted (56 % measured), Laikipia 52 % (52 %). A measured shrinkage curve or COLE would test the model directly.